

of space, thanks to four exterior mounting platforms that can accommodate external payloads. With a clear view of Earth and the vastness of space, external experiments can run the gamut from the microscopic world of bacteria to the limitlessness of space.

Dextre

Along with the Canadarm2, the country has also contributed what is popularly known as the ‘Canada Hand’. Dextre – installed in March 2008 – is a two-armed robotic device used for servicing around the space station. The Special Purpose Dexterous Manipulator, or Dextre, is capable of handling the delicate assembly tasks usually handled by astronauts during space walks. It has built-in grasping jaws, a retractable socket drive, a monochrome TV camera, lights, and an umbilical connector that can provide power, data, and video to/from a payload.

Kibo

The same Space Shuttle Endeavour that carried Dextre also had Japan’s first major contribution to the ISS. JAXA’s Experiment Module called Kibo (Japanese for “hope”) is Japan’s first human space facility and enhances the unique research capabilities of the ISS. Experiments focus on space medicine, biology, Earth observations, material production, biotechnology and communications research. The Pressurised Module research facility – about the size of a tour-bus – is a shirt-sleeve environment in which astronauts conduct microgravity experiments. The Exposed Facility is a unique platform on the ISS that is located outside of the Pressurized Module and is continuously exposed to the space environment. The platform can hold up to 10 experiment payloads at a time and measures 5.6 metres wide, 5 meters high and 4 metres long. Items positioned on the exterior platform focus on Earth observation as well as communication, scientific, engineering and materials science experiments. Astronauts exchange experiment payloads or hardware from the Pressurized Module through the scientific airlock using the Kibo Remote Manipulator System (RMS). The RMS consists of two robotic arms – Main Arm and the Small Fine Arm for delicate operations – that support operations on the outside.

Poisk

The Russian Mini-Research Module 2, Poisk (Russian for “search”, “seek” or “explore”) docked to the space-facing port of the Zvezda service module on November 12, 2009. It provides an additional docking port for visiting